Sysc 4907-E - 4th Year Capstone Project Proposal

Edge-AI Home Anomaly Detection for Elder Care

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1.1 Problem Definition & Background

1.1.1 Problem Definition

Older individuals living independently may experience falls or medical emergencies at home with no one present to assist. Many emergency-response solutions require wearing a device or manually activating an alarm. If the individual becomes unconscious, immobilized, or disoriented, or is not wearing the device, an emergency may go undetected.

This project addresses the need for a passive, privacy-preserving system that monitors in-home activity and identifies anomalous behaviour that may indicate a fall or medical emergency, without wearables or cameras.

1.1.2 Background and Importance

Falls seriously threaten older adults' health and safety. The Public Health Agency of Canada identifies falls as the leading cause of injury-related hospitalization and death among Canadians aged 65+. Falls can cause disability, reduced mobility and independence, long-term care admission, and death [1]. Such deaths rose 51% from 2017 to 2022 [2].

This especially affects those living alone. In 2021, 32% of Canadians aged 80–84 and 42% aged 85+ lived alone [3]. As Canada ages, technologies supporting safe, independent living are increasingly needed.

1.1.3 Current Gap and Limitations

Existing approaches involve trade-offs between reliability, usability, privacy, and cost. Personal emergency response systems and wearables help, but only if the individual has, wears, and can activate the device.

Fall-detection research spans wearable and non-wearable technologies, but accuracy, real-world validation, user acceptance, and privacy remain challenges [4]. Cameras offer detail but raise privacy concerns and are out of scope.

This motivates the development of a low-cost, passive, camera-free monitoring system using acoustic sensing and edge AI to detect potential falls and unusual household activity. By utilizing affordable, commercially available hardware and local processing, the system aims to provide an accessible and privacy-preserving solution. This project investigates whether a home's acoustic fingerprint can be learned and used to detect falls and anomalous behaviours that may signal danger to an independently living senior, without requiring them to request help.

1.2 Existing Solutions (State-of-the-Art Review)

Various technologies aim to improve older adults' safety at home, including wearable alerts, camera monitoring, radar, and smart-home activity analysis. Each solution trades off one or more of these characteristics: detection capability, privacy, user interaction, deployment complexity, and cost.

1.2.1 Lifeline HomeSafe with Fall Detection

Lifeline HomeSafe is a commercial wearable in-home emergency response system whose pendant detects falls and can contact a 24/7 response centre even if the user cannot press the button [5].

It offers mature emergency response but requires continuous wear; if removed or forgotten, protection is reduced. It also targets discrete falls rather than broader behavioural changes.

1.2.2 Nobi Smart Lamp

Nobi is a commercial ceiling-mounted smart light for older adults and care facilities. It uses AI and image-based sensing to detect falls, monitor movement, recognize unusual behaviour, and notify caregivers [6].

Nobi is passive, wearable-free, and processes images locally, but relies on visual sensing, excluded here for privacy.

1.2.3 Vayyar Care

Vayyar Care is a commercial system using radio-frequency sensing instead of cameras or wearables. It provides fall detection, movement monitoring, and alerts for unusual movement [7], showing passive monitoring works without visual or audio surveillance.

Though among the closest solutions to this project, it is proprietary, relies on specialized radar hardware, targets professional senior-care settings, and requires a post-installation learning period [8]. These factors can increase cost and deployment complexity compared with a low-cost, flexible prototype.

1.2.4 mmFall

mmFall is an academic system combining 4-D millimeter-wave radar with a Hybrid Variational Recurrent Neural Network Autoencoder. A rising anomaly level combined with decreasing body height indicates a potential fall [9].

It is passive, camera-free, and open-source, making it highly relevant. However, mmWave radar and recurrent networks add significant hardware, signal-processing, and machine-learning complexity, and it targets only falls.

1.2.5 Smart-Home Activity Anomaly Detection

Shahid, Saguna, and Åhlund developed a smart-home system using IoT sensors in older adults' residences. It learned activity duration and frequency, identified deviations from normal routine, and notified relatives via SMS [10].

This approach is highly relevant because it targets behavioural change rather than only falls. However, it requires sufficient historical data to establish a routine, and multiple sensors complicate installation and configuration.

1.2.6 WiFi CSI-Based Fall Detection

WiFi Channel State Information (CSI) can detect activity without cameras or wearables. A 2025 study's lightweight convolutional (CNN) neural network identified falls from WiFi signal changes with high accuracy in controlled environments [11].

WiFi sensing may reuse existing infrastructure and preserves visual privacy. However, accuracy dropped with unseen environments and individuals, showing generalization to real homes is difficult.

1.2.7 Acoustic-FADE

Acoustic-FADE is an academic fall-detection system developed by Li, Ho, and Popescu. It uses a circular microphone array to capture sounds within a room, estimate their location, and classify detected events as falls or non-falls. The system also applies signal-processing techniques to improve detection in noisy environments and reduce false alarms [12].

A key advantage of Acoustic-FADE is its ability to detect falls passively without requiring wearable devices or visual surveillance. However, its microphone-array configuration and sound-localization methods introduce additional processing complexity. The system also primarily focuses on identifying falls rather than monitoring longer-term behavioural changes.

1.2.8 Neo Care

Neo Care is a commercial sound-based home-monitoring system that uses artificial intelligence to detect potentially concerning household sounds, including falls, cries, and breaking glass.

When a concerning event is detected, the system can interact with the resident and contact a caregiver when assistance may be needed [13].

A 2024 study evaluated Neo Care over 12 weeks in 18 households with older adults. Although the system demonstrated the potential of passive, wearable-free monitoring, it frequently responded to ordinary household sounds and missed some falls. Connectivity problems also affected its operation. These findings highlight the challenges of distinguishing genuine emergencies from harmless sounds and maintaining reliable monitoring in real residential environments [13].

1.2.9 Comparison of Existing Solutions

To summarize the state-of-the-art review, Table 1.1 compares the key characteristics, advantages, and limitations of the existing commercial and academic solutions discussed above. The comparison highlights the trade-offs between privacy, usability, hardware complexity, cost, and detection capability across the different approaches.

Solution	Key Characteristics / Needs	Limitations & Constraints
Lifeline HomeSafe (Commercial)	Wearable pendant, automatic fall detection, 24/7 response centre, two-way communication. Mature system and widely deployed emergency response.	Requires user to wear device; mainly detects falls rather than behavioural anomalies; medium cost (commercial service & hardware).
Nobi Smart Lamp (Commercial)	AI-based fall detection, activity monitoring, caregiver alerts, smart lighting. Passive monitoring with local image processing and no wearable required.	Uses visual/image sensing (incompatible with camera-free requirement); high cost (dedicated smart-light hardware, installation, commercial platform).
Vayyar Care (Commercial)	RF/mmWave-style sensing, fall detection, unusual-movement monitoring, automated alerts.	Proprietary, specialized hardware with post-installation learning period; aimed largely at professional care

Solution	Key Characteristics / Needs	Limitations & Constraints
	Passive, privacy-preserving sensing with no camera or wearable.	environments; high cost and integration complexity.
mmFall (Academic / Open Source)	4-D mmWave radar, body-point-cloud processing, RNN autoencoder, anomaly-based fall detection. Camera-free, passive, strong privacy, open-source.	Specialized radar and complex signal processing/ML; primarily fall-focused; high hardware and processing complexity.
Smart-Home Activity Anomaly Detection (Academic)	Multiple IoT sensors, routine learning, statistical anomaly detection, caregiver SMS notifications. Detects broader behavioural changes passively in real homes.	Requires multiple sensors and sufficient historical data to establish normal routines; medium–high setup and modelling complexity.
WiFi CSI Fall Detection (Academic)	Uses WiFi signal changes with lightweight CNN to identify human activity and falls. No wearable or camera required; potentially reuses existing wireless infrastructure.	Performance varies between homes and individuals; environmental changes affect signals; requires CSI-capable hardware/software and ML processing.
Acoustic-FADE (Academic)	Circular microphone array, sound localization, signal enhancement, automatic fall detection. Camera-free and wearable-free.	Specialized array and signal-processing complexity. Primarily detects falls instead of focusing on longer-term behavioural changes.

Solution	Key Characteristics / Needs	Limitations & Constraints
Neo Care (Commercial)	AI-based acoustic monitoring, detection of concerning sounds, voice interaction, caregiver alerts. Passive and wearable-free.	High false-activation frequency in the evaluated homes, missed falls, and connectivity issues.

Table 1.1: *Comparison of Existing Solutions*

1.2.10 State-of-the-Art Gap

Existing solutions demonstrate the feasibility of automatic monitoring, but each presents trade-offs in reliability, privacy, usability, cost, and deployment complexity. The most suitable approach depends on balancing these factors to meet the needs of older adults living independently. Wearables enable reliable emergency communication but require compliance. Cameras provide detail but raise privacy concerns incompatible with this project. Radar and WiFi improve privacy but may need specialized hardware, complex processing, or per-environment tuning. Behavioral smart-home models detect broader anomalies but need many sensors and enough routine data.

Acoustic-based approaches such as Acoustic-FADE and Neo Care demonstrate the feasibility of detecting falls using microphones without requiring wearable devices or cameras. However, these approaches also highlight limitations involving false alarms, environmental noise, and reliable detection in real-world conditions. Furthermore, both systems primarily focus on detecting individual events rather than learning longer-term household behavioural patterns [12], [13].

This project therefore investigates a low-cost, passive, acoustic-based edge-AI monitoring system that aims to balance these trade-offs through the following features:

- **Passive, wearable-free monitoring:** Uses microphones to monitor household activity without requiring the resident to wear or activate a device.
- **Privacy-preserving design:** Operates without cameras and processes audio locally on the edge device without transmitting raw audio.
- **Fall detection:** Identifies acoustic patterns associated with potential fall events.
- **Behavioural anomaly detection:** Learns normal household acoustic patterns and identifies significant deviations that may indicate danger.

- **Reduced false alerts:** Classifies detected events and considers contextual information to help distinguish potentially dangerous situations from normal household activity.
- **Caregiver notifications:** Provides alerts through email and a web dashboard displaying event history and device status.

Together, these features aim to provide affordable, privacy-conscious monitoring that can identify both immediate fall events and broader behavioural anomalies while requiring minimal user interaction.

1.3 Stakeholders / Clients

The proposed system involves several stakeholder groups with different needs, responsibilities, and constraints.

1.3.1 Primary Clients & Deployment Stakeholders

- **Family Caregivers (Including Adult Children):** Family caregivers are potential purchasers of the system and the primary recipients of emergency alerts.
    - **Needs:** Timely and reliable notifications, sufficient information to assess potentially dangerous situations, and reassurance without needing to constantly monitor the system.
    - **Constraints:** Limited availability, geographical distance from the senior, and the emotional and practical demands of caregiving [14].
    - **Design Implications:** Alerts should provide relevant context, including event type, room location, and timestamp. False alarms must be minimized to avoid alert fatigue, while a web dashboard should provide access to alert history and device status.
- **Home-Care Services and Retirement Communities:** These organizations are potential clients that may deploy the system to support the monitoring of older adults.
    - **Needs:** Affordable and straightforward installation, reliable detection, minimal unnecessary alerts, and protection of residents' privacy.
    - **Constraints:** Limited budgets, varying levels of technical expertise, and different residential environments.
    - **Design Implications:** The system should use affordable hardware, require minimal maintenance, and provide actionable alerts.

In addition to the primary clients, end users, and family caregivers, several secondary stakeholders may influence the design and deployment of the proposed system. Table 1.2 summarizes the key needs of these stakeholders and the corresponding constraints they impose on the system.

Secondary stakeholder	Need	Constraint imposed
PSWs / professional caregivers	Know quickly if a client is in trouble between visits	Alerts must be actionable and prioritized; no added documentation burden
Home-care agencies / health authorities	Fewer avoidable transports and premature LTC admissions	Auditable event logs and measurable performance metrics
Emergency responders	Accurate dispatch; no false dispatches	Escalate to human confirmation rather than auto-dialling 911
Installers / maintainers	Fast, repeatable install in arbitrary homes	Self-calibration; remote device health monitoring without audio access
Health Canada	Whether the system is a medical device	Out of scope for this prototype (see Section 1.6); the system is presented as a safety notification prototype, not a medical device
Other household members and visitors	Not to be analyzed without consent	Consent and disclosure must extend beyond the primary resident

Table 1.2: *Secondary Stakeholder Needs and Constraints*

1.3.2 End Users (Monitored Seniors)

The primary end users are older adults living independently. The system is intended to operate passively while allowing the monitored senior to control when monitoring is active.

- **Needs:** Independence, timely assistance during potential emergencies, minimal disruption to daily life, and control over monitoring. Privacy is particularly important, as users must feel confident that the system does not expose their private conversations or household activities.
- **Constraints:** Limited technical skills, possible sensory or cognitive impairments, reluctance to wear or charge devices, unreliable Wi-Fi connectivity, and fixed budgets.
- **Design Implications:** The system should require no routine interaction after installation, not use cameras or wearable devices, and provide an accessible physical mute control on the device. All raw audio must remain on the local edge device, with only event information transmitted to the caregiver application.

1.4 Objectives

The main goal of this project is to design, build, and evaluate an edge-AI home-monitoring system capable of detecting falls and anomalous household activity that may indicate a potentially dangerous situation for an older adult living independently. This is achieved by focusing on acoustic sensing and local processing to avoid the use of cameras and wearable devices in order to respect privacy.

1.4.1 Primary Objectives

- To design a passive monitoring system that captures household sounds using microphones without requiring any wearable devices.
- Develop and evaluate a fall-detection method that can identify acoustic patterns associated with potential fall events on the edge device.
- Investigate behavioural anomaly detection methods that learn normal household activity patterns and identify significant deviations that may indicate risk.
- Distinguish potentially dangerous events, such as falls, from harmless household sounds, such as dropped objects, by combining sound classification, anomaly scores, and contextual information to reduce unnecessary alerts.
- Evaluate the system's detection performance under different microphone placements, distances, and background noise conditions, such as television sounds, running appliances, and other household activity. Report detection accuracy, missed events, and false alerts.
- Investigate whether combining acoustic fall detection with household activity patterns and anomaly scores reduces false alerts compared with using acoustic fall detection alone.
- Minimize processing and communication latency to enable timely delivery of caregiver alerts.

- To build an alerting system that would notify a caregiver or an emergency contact person of the situation.
- Preserve the senior's privacy by not using video surveillance, processing all audio locally, and ensuring that raw audio is never transmitted off the edge device.
- Create a prototype by incorporating sensing, processing, detection, and alert systems to test the system as a whole.
- Design an intuitive web interface for viewing the alerts and checking the system status.

1.4.2 Secondary Objectives

- Investigate methods for adapting the learned acoustic baseline to long-term changes in household routines, such as changes in daily schedules, appliance usage, and activity patterns.
- Explore ways to improve the system's generalization across different household environments, including variations in room layouts, background sounds, and occupants' daily routines.
- Analyze the possibility of scaling the system to support multiple rooms, additional microphones or other sensor types, and multiple occupants.

1.5 Requirements

1.5.1 Functional Requirements

FR1 - Audio Sensing: The system shall continuously capture audio using one or more microphones placed in a monitored room.

FR2 - Local Processing: The system shall process all audio on the embedded device. Raw audio may be stored locally for a maximum of 24 hours before being automatically deleted and shall never be transmitted off-device.

FR3 - Fall Event Detection: The system shall detect acoustic patterns associated with potential fall events.

FR4 - Baseline Learning: The system shall learn a baseline of routine household acoustic activity, including common sounds (e.g., running water, appliances, footsteps), their typical durations, and their frequency of occurrence over defined observation periods.

FR5 - Anomaly Detection: The system shall identify potential anomalies when acoustic patterns, sound durations, or frequencies of occurrence exceed predefined thresholds established from the learned baseline of normal household activity.

FR6 - Event Data Generation: On detecting a fall or anomaly, the system shall generate an event record containing event type, room, timestamp, and elapsed time since detection.

FR7 - Event Classification: The system shall assess detected events using sound classification confidence, anomaly scores, and temporal information to distinguish low-risk household events from potentially dangerous situations. Events exceeding a predefined risk threshold shall trigger caregiver alerts, while events below this threshold shall not.

FR8 - Alert Transmission: The system shall transmit event data (not raw audio) to a cloud server, which shall display the alert on the web dashboard and notify the caregiver by email through a free email service.

FR9 - Dashboard Reporting: The system shall provide a web dashboard displaying alert history and device status, including network connectivity, whether monitoring is active or muted, and the device's last communication time.

FR10 - Manual Mute: The system shall provide a physical mute control that immediately stops audio capture and processing.

FR11 - Monitoring Status Indicator: The system shall provide a visible indicator showing whether audio monitoring is currently active or muted.

FR12 - Outage Handling: The system shall continue local detection during a network outage and queue or retry alert delivery once connectivity is restored. Delayed notifications shall be clearly identified as such and include the original event detection timestamp.

1.5.2 Non-Functional Requirements

NF1 - Detection Latency: The system shall create a fall event record within 10 seconds of the fall sound. Most of this time is the short check for silence after the impact (Section 1.7.4). The model itself shall process each 0.96-second audio window in under 200 ms, so the device keeps up with live audio. Behavioural anomalies build up over minutes or hours, so they shall be flagged within 5 minutes of the deviation crossing its threshold.

NF2 - Alert Latency: Once an event record is created, the alert shall appear on the web dashboard within 10 seconds, and the email shall be handed to the email service within 30 seconds. This means a caregiver can see a fall alert on the dashboard within about 20 seconds of the fall. After a network outage, queued alerts shall be sent within 60 seconds of the connection returning.

NF3 - Detection Performance: Overall accuracy alone can hide a high number of missed emergencies or false alarms. Results shall therefore report overall accuracy, missed events (false negatives), and false alerts (false positives) separately. Targets:

- Fall detection on the held-out SAFE test set: at least 90% overall accuracy, no more than 10% of falls missed, and no more than 5% of non-fall sounds flagged as falls.
- Staged falls (weighted dummy and mats, in a room the model was not trained on): no more than 20% of falls missed.

NF4 - Reliability: The system shall meet these targets across varying levels of household activity and background noise. It shall also run continuously for at least 7 days, the length of the SINS recordings, without crashing or needing a manual restart.

NF5 - Memory and Resource Use: On the Raspberry Pi 5 (2 GB), the full detection pipeline (audio capture, fall model, baseline and event queue) shall use no more than 1 GB of RAM and no more than 50% of the total four-core CPU capacity on average. This leaves headroom for the operating system. The deployed models shall take up less than 20 MB. Stored audio shall be kept in a rolling buffer capped at 64 GB of the 128 GB microSD card, with recordings automatically deleted after 24 hours or earlier if the storage limit is reached.

NF6 - Power Consumption: The device (Raspberry Pi and microphone array) shall draw no more than 7 W on average and 12 W at peak, measured with a USB-C power meter over 24 hours of continuous operation. This stays well within the 27 W power supply. The processor shall stay below 80 °C so it does not throttle, which supports the thermal considerations in Section 1.11.

NF7 - Cost: The hardware shall cost no more than $500 CAD in total, including tax and shipping. It shall use low-cost, commercially available development hardware and open-source software, consistent with the course's component-return requirements. The required parts are currently estimated at about $335. Cloud services shall run on free tiers (Azure for Students and a free email service), so the recurring cost during the project is $0 per month.

NF8 - Usability: The system shall require zero post-installation interaction from the end user, with no wearables, no manual activation, and a simple mute/indicator interface.

NF9 - Safety: The system shall not interfere with other household electronics, shall have no user-facing electrical or mechanical hazards, and shall not autonomously contact emergency services (911), instead escalating to human confirmation.

NF10 - Privacy: The system shall never transmit raw audio off the device, and any stored audio shall remain on the local device. All sensitive processing shall occur on the local device, consistent with the privacy and consent considerations defined in Section 1.12.

NF11 - Scalability: The architecture should be extensible to additional rooms, sensors, or occupants, even though multi-room/multi-resident support is out of scope for this prototype (per Section 1.6).

1.6 Scope Definition

Section 1.4 establishes the goal of detecting falls and anomalous household activity that may indicate danger to seniors living alone. This section narrows that goal to what can be built and tested in two terms. It keeps the work focused on the project's central research question: whether a home's acoustic fingerprint can be learned on embedded hardware and used to detect falls and anomalous behaviours accurately.

1.6.1 Included Scope

- Microphones are the only sensors, and all audio is processed on the embedded device. Raw audio may be stored on the device but is never transmitted. Only event data leaves the device: the event type, room, time, and elapsed time. The primary detection targets are potential fall events and anomalous household activity. Fall events will be identified through acoustic patterns, while behavioural anomalies will be identified through deviations from learned normal household activity. Detected events will then be classified by type and risk so that low-risk events do not produce unnecessary caregiver alerts.
- The device will learn a baseline of routine household acoustic activity, including common sounds such as running water, appliances, and footsteps, along with their typical durations, frequencies, and timing patterns. The prototype will evaluate a limited set of temporal anomalies, including prolonged running water, unusually frequent impact sounds within a defined time window, and a fall-like impact followed by an extended period of reduced household sound activity. These patterns will be evaluated against thresholds established from the learned baseline, with prolonged silence treated as supporting evidence rather than confirmation of an emergency. Because the prototype will not be deployed in a real home, the baseline will be learned from the SINS database, which contains continuous audio recordings of one person living in a vacation home over one week, labelled with daily activities such as cooking and showering [15]. Recordings from a single room will be replayed through the device's processing pipeline in time order, so the baseline is learned the same way it would be in a home. The non-fall recordings in SAFE [16] will provide further examples of normal sounds.
- The anomaly-detection results also assume that the SINS recordings are close enough to a senior's household to be informative. SINS covers one person in one home over one week, so the results will show whether the method works, not how well it generalizes to other homes or longer periods.

- Alerts are sent as event data to a cloud server, which shows them on a simple web dashboard and emails the caregiver through a free email service. The dashboard also shows alert history and device status. As identified in Section 1.3, the device includes a physical mute control and a visible indicator showing whether audio monitoring is active or muted.
- Testing combines staged falls with public datasets. The staged falls use a weighted dummy and controlled falls onto mats. The main dataset is SAFE, which contains 950 audio samples, including 475 simulated fall recordings using a dummy [16]. Results will be reported as overall accuracy, missed falls, false alerts, and detection and alert latency, and compared against the targets in Section 1.5.

1.6.2 Excluded Scope

- Cameras, wearables, and other non-visual sensors such as radar and WiFi CSI are excluded. Other sensing methods are considered only in the scaling analysis and future work in Section 1.9. The system will not transcribe or interpret speech.
- The project will not integrate with nurse-call platforms. This need is deferred to future work (Section 1.9). The system will also not place automatic calls to 911, which Section 1.3 already rules out in favour of human confirmation.
- The prototype covers one room and one occupant. Whole-home and multi-resident coverage will be analyzed but not built. Because only one room is monitored, the room field in each event record will always be the same, so the senior's need for fast location during emergencies (Section 1.3) is only partly met and is deferred along with multi-room support. The system flags events that may need attention and does not diagnose medical conditions.
- Trials with older adults or other volunteers, and long-term deployment in real homes, are also excluded. TCPS 2 requires Research Ethics Board approval before any research involving living human participants begins, and this includes course-based projects [17].
- Formal Health Canada medical-device classification is outside the scope of this project. Whether software is regulated as a medical device depends largely on its intended use and how it is represented [18].

1.6.3 Assumptions and Constraints

- The design assumes a single older adult living alone, the group highlighted in Section 1.1 [3]. Visitors will appear as deviations from the baseline, which is accepted as a limitation.
- It also assumes that staged falls sound close enough to real falls to be informative. The SAFE authors caution that a dummy may not sound like a real person falling [16]. Results will therefore show feasibility under controlled conditions rather than real-world accuracy.

- Detection is assumed to run without a network connection, while alert delivery requires one. Behaviour during outages will be tested under the reliability objective in Section 1.4.

The project runs from Fall 2026 to Winter 2027, and the integrated prototype must be ready for the poster fair demonstration in late March 2027. The supervisor allocates the budget, and all purchases go through the department's approval process, with components returned at the end of the course. This favours low-cost, commercially available development hardware and open-source audio and machine-learning tools. The team's size and undergraduate experience are the main reason the alert side is limited to email notifications and a basic dashboard.

1.7 Proposed Solution (High-Level Design)

1.7.1 System Overview & Design Philosophy

The proposed system is an edge based acoustic monitoring system intended to monitor and detect anomalous sounds in a home environment. The sounds that the system detects as anomalies are potentially concerning for the wellbeing of an elderly adult that is living independently. The system aims to provide passive monitoring through acoustic observations while learning the household's normal sound activity. The primary design objectives are to detect potentially concerning acoustic events, establish a baseline of normal household activity, preserve the privacy of captured audio, and notify designated caregivers when an event warrants attention.

To accomplish these objectives, the system is divided into three primary technical domains: Edge Hardware, Edge Software, and External Software.

Edge Hardware:

- Microphone(s) responsible for capturing acoustic signals
- The edge computing device responsible for local processing.

Edge software components:

- **Audio preprocessing:** Responsible for ingesting and transforming the digital audio stream into short spectrogram segments for machine learning analysis.
- **Machine learning:** Responsible for general sound classification generating learned sound vector embeddings. The embeddings are used in fall detection by a specialized classification layer and.
- **Specialized sound classifier layer:** An additional layer that takes the output of the general open source sound classifier to classify specific sounds like falls.
- **Anomaly detection:** This component will also use vector embeddings of sound to construct a statistical model of household baseline further expanded in section 1.7.3.2.

- **Event Manager:** Component where the sound classification and anomaly detection will be analyzed and created into an event if they indicate a concern. Also responsible for sending events to the external web application.

External software components:

- **Frontend:** Client side web dashboard that provides caregivers a view of detected events and received notifications.
- **Backend:** Server side application receives and manages events, communicates with edge device and email notification service.
- **Email Notification Service:** Recess events from application and send email notification to respective caregiver.
- **Database:** Persist user event data for access on the web dashboard along with accounts and credentials, and lastly overall system logs.

1.7.1.1 Architecture Overview

Below figure 1.1 illustrates the system at a high level across technical domains and home environment. The majority of the system will operate within the residence of an elderly adult outlined by the home environment. Within the home environment the three technical domains converge starting with hardware, acoustic signals of the home environment will be captured as input via the microphone(s). The audio data will be transmitted to the main edge device for processing, analysis, and potential detection.

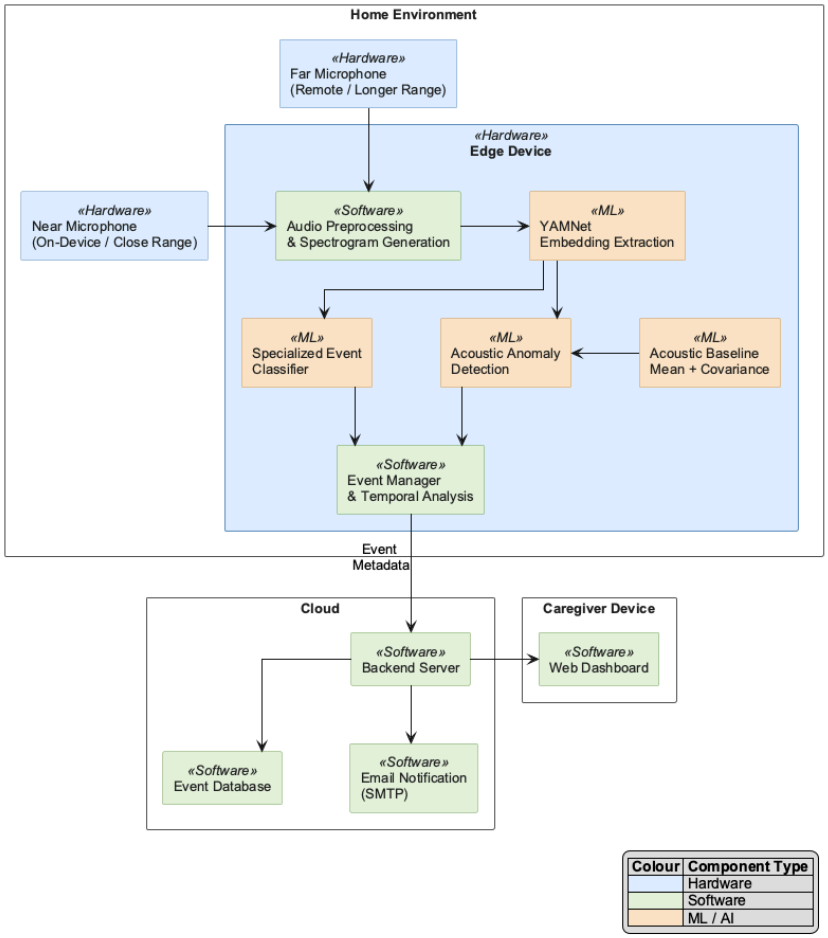


Figure 1.1: *High level Architectural Block Diagram*

1.7.1.2 Audio Preprocessing Overview

The audio preprocessing component will preprocess the audio for further analysis. The preprocess step will take in the captured digital audio stream, perform normalization and divide the continuous stream into short overlapping time segments. These segments will be computed into spectrograms which represent frequency over time. The spectrogram extracted from the raw audio segments will be the input to a pretrained audio classifier on the same edge device.

1.7.1.3 Machine Learning Overview

Leveraging a pre-trained audio model reduces the computational resources, training data, and development time required to construct an audio feature extraction model from scratch. The pretrained model will generate learned feature embeddings that numerically represent the acoustic characteristics of the captured audio. These embeddings will support two complementary detection processes: event classification and acoustic anomaly detection. Event

classification will identify sounds associated with predefined events, such as potential falls, while anomaly detection will identify deviations from a locally maintained statistical baseline of household acoustic activity.

1.7.1.4 Event Manager & Notification Overview

The Event Manager will evaluate the classification results and anomaly scores to determine whether an observation warrants generating an event. Detected events will contain relevant metadata, such as the event type, timestamp, and confidence or anomaly score, while excluding raw audio data. This approach ensures that sensitive audio remains within the home environment and only information necessary for caregiver notification is transmitted externally.

When an event is generated, it will be transmitted to a cloud-hosted server through a secure network connection. The backend application will record the event in a database and initiate an email notification through a Simple Mail Transfer Protocol (SMTP) service. The event information will also be made available through a web dashboard accessible to designated caregivers. These notifications are intended to inform caregivers of potentially concerning acoustic events that may require attention.

1.7.1.5 System Summary & Philosophy

The system follows a privacy-focused, edge-based design philosophy in which audio processing and event detection occur locally without requiring continuous transmission of raw audio to an external server. By combining general-purpose audio classification with a personalized acoustic baseline, the system aims to identify potentially concerning events while accommodating differences in household sound environments.

1.7.2 Audio Acquisition & Feature Extraction

Audio acquisition is done continuously by two microphones with wired connection to the Edge computing device. A small microphone will be located at the device and a larger microphone can be placed further from the device to capture larger space. Having two microphones improves audio capture redundancy and allows audio from different vantage points in the room, especially a large space. Now two audio streams can be configured in the same large space or expand to another room. Having two microphones ensures reliability by having at least one microphone monitoring while the system is operational. Other factors that may affect the quality of the captured sound are how far a sound is from the microphones, background noise, and overall room layout. Multiple configurations will be tested to asset the systems capability. Each audio stream will be processed independently through the pre-trained audio classifier. The audio captured will be single-channel (mono) audio sampled at 16 kHz to maintain compatibility with the pre-trained audio model.

The audio preprocessing component will prepare the captured digital audio stream for machine learning analysis. This process will include normalization and segmentation of the continuous stream into short, overlapping audio segments. Each segment will be transformed into a log-mel spectrogram using the Short-Time Fourier Transform (STFT) and mel-frequency filtering. The resulting spectrogram provides a time-frequency representation of the captured sound.

The proposed system will use YAMNet, a pretrained audio classification model developed by Google, as the foundation for acoustic feature extraction and sound classification. YAMNet employs the computationally efficient MobileNetV1 architecture and has been trained to recognize 521 categories of environmental sounds using the AudioSet dataset [19]. YAMNet's computationally efficient design is an advantage as this will be deployed to the edge device that has limited computational resources.

The primary advantage of leveraging YAMNet is its ability to extract learned acoustic representations without requiring the development and training of a neural network from scratch. The model produces 1,024-dimensional feature embeddings that can be used as inputs to downstream detection components. These embeddings will support both a specialized classifier for identifying potential fall-related sounds and an acoustic anomaly detector that compares new observations against a baseline of normal household activity. Additional advantages of YAMNet is it comes with feature extraction support to compute the spectrogram via STFT in the preprocessing step. Leveraging existing implementations of complex computation will reduce error and ease development.

The figure 1.2 below represents the data transformation described above and transmission from digital audio streams to JSON representation of events for the web application.

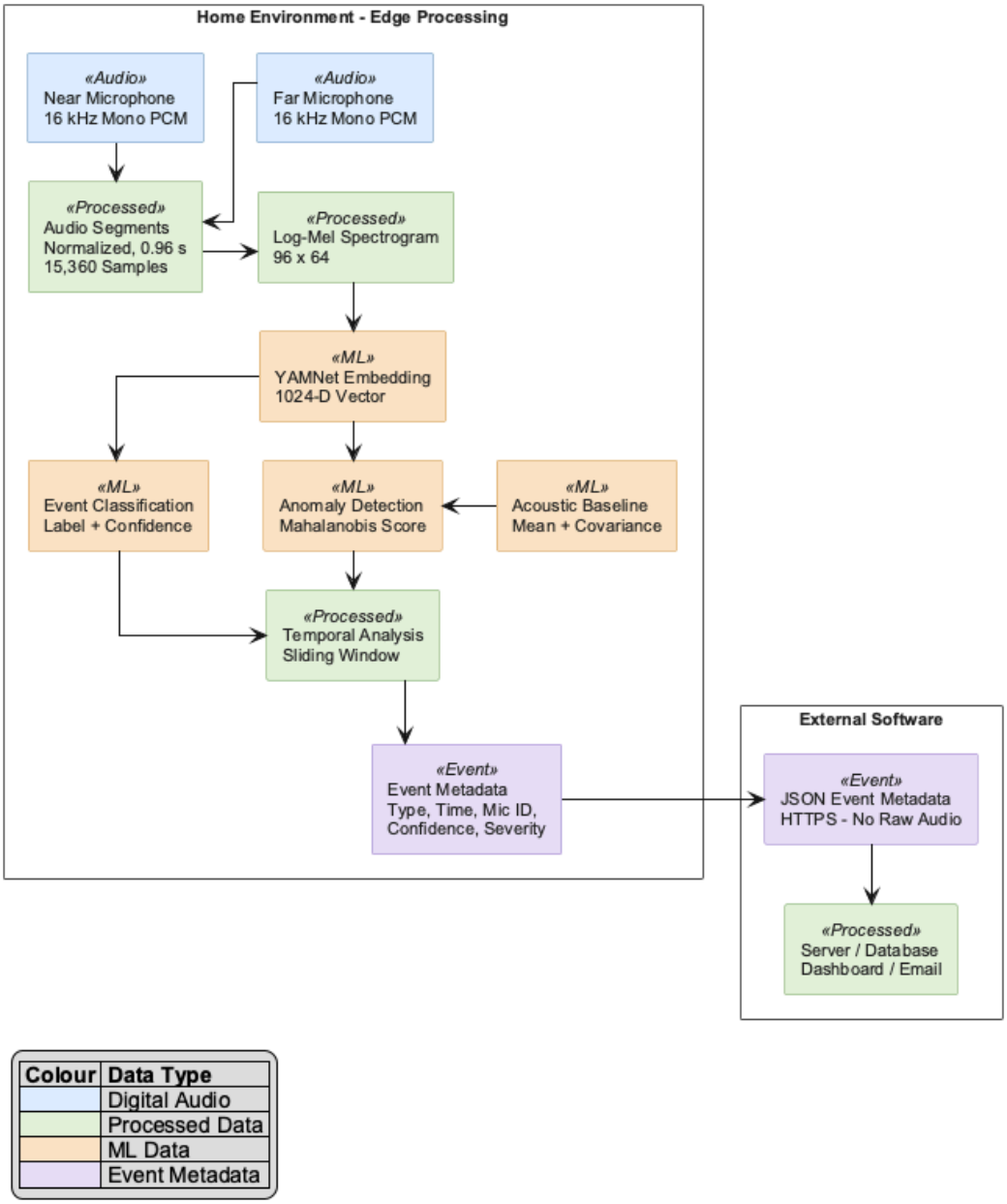


***Figure 1.2**: Data Transformation - Audio to Event*

1.7.3 Acoustic Event Classification & Anomaly Detection

1.7.3.1 Acoustic Event Classification

YAMNet only serves as the foundation for feature extraction as it is a general sound classifier that has a large learned set of sounds. To refine the classification, for the purposes of identifying potential concerns like a fall or glass breaking an additional specialized classifier will be

implemented. A similar approach to this is this TensorFlow tutorial: “Transfer Learning with YAMNet for Environmental Sound Classification” will be employed. In the tutorial an additional classifier is developed from YAMNet embeddings that focuses on dog and cat sounds using embeddings extracted from the ESC-50 dataset [20]. For the purpose of the proposed system the tutorial’s approach can be replicated with the SAFE (Sound Analysis for Fall Events) Dataset mentioned above in section 1.6 Included Scope to produce a light weight classifier able to distinguish falls. The SAFE dataset 475 simulated fall events can be passed through YAMNet and the resulting embeddings can be used to train a fall-like sound classifier. Additional datasets may be incorporated to support the classification of other potentially concerning household sounds. For training the specialist classifier, fall and non-fall samples of the SAFE data set will be used. To assess its capabilities of distinguishing the falls from the regular impacts at home, extra non-fall audio samples, including the falling objects and moving pieces of furniture, will be added for controlled testing, if possible. Instead of recognizing every impact as a fall, the classifier will return a fall-confidence score for the Event Manager to consider together with the anomaly score and time information to avoid false alarms.

1.7.3.2 Anomaly Detection

Anomaly Detection will also utilize the YAMNet Embeddings. To accomplish anomaly detection it is essential to establish a normal acoustic baseline of the home. This can be accomplished by constructing the baseline using statistical methods to characterize typical household sounds and patterns of acoustic activity.

The current favoured approach is inspired by Detection and Classification of Acoustic Scene and Events (DCASE) challenge Task 2. DCASE 2026 Task 2 titled: “Noise-Aware Unsupervised Anomalous Sound Detection for Machine Condition” (Noise-Aware UASD) uses two approaches to anomaly detection [21]. A Simple Autoencoder which uses reconstruction error and Selective Mahalanobis distance scoring that incorporates covariance statistics into reconstruction-error scoring [22]. The proposed system will focus on the second approach of computing a Mahalanobis distance of a new embedding against the learned Baseline represented by the mean embedding and its covariance matrix accounting for the variation between embeddings.

These three acoustic anomalies are fall-like sounds, prolonged running water, and abnormal repeated impacts. Fall-like sounds will refer to all kinds of impacts, other than those from the baseline, that occur in the environment and will be compared against the most common house noises such as fallen objects or movement of furniture. The condition of prolonged running water will be determined when the duration of running water sounds is significantly higher than the expected durations. Abnormal repeated impacts will be determined based on higher frequencies of impacts than expected in an observation period.

Sound anomalies will be detected based on the Mahalanobis distance between the YAMNet embedding vectors of new samples and the learned baseline. However, some anomalies are based on sound duration or frequency of sounds, and therefore, a more lengthy analysis is needed. In this case, the baseline will include additional statistics regarding regular durations and frequencies of sounds. Then the Event Manager will analyze the duration of running water sounds and the number of impacts using sliding time windows in comparison with the baseline.

1.7.3.3 Baseline Representation & Management

To establish this baseline a YAMNet embeddings are collected over a calibration period. Once there is sufficient number of embeddings the baseline is computed and new embeddings can be scored with Mahalanobis distance from the baseline. Far from the mean, the sound will be considered anomalous.

In relation to the prototype capstone, the main aim would be setting up of a static baseline at the very beginning of the calibration process for the SINS data set as highlighted in section 1.6. The baseline will always be static through the process of evaluation in order to maintain consistency in the process of detection and to avoid any kind of unusual and possibly dangerous situation being included in normal behavior automatically. One of the possible improvements that may be done in the future is recalibrating the baseline from time to time due to any kind of changes in household activities or even with the addition of new home appliances.

1.7.4 Event Management & Decision Logic

At this stage in the pipeline the Event Manager will receive two inputs from a single YAMNet embedding: a specialized audio classification i.e. fall or glass break and an anomaly score derived from the Mahalanobis distance. The combination of these two outputs will determine if an event is sent to the caregiver application. Combination of classified sound like fall and high anomaly score can result in the creation of an event. However both metrics do not have to be high to create an event. Perhaps there are a lot of thud sounds from a mobility aid like a cane that will be part of the baseline. A new thud might get classified and a dish or object falling but produces a low-normal anomaly score. To still alert of this detection event creation will favour the sound classification component for event creation by weighted sum of the two scores.

To handle temporal events that occur over time the event manager will track and keep a sliding window of sound classifications and anomaly scores. This enables the system to identify patterns based on duration, frequency count of sounds, and sequence of sounds.

Silence for extended periods cannot confirm physical inactivity or emergencies as the resident could simply be asleep, reading, or not in the room under observation. In that case, silence will only serve to provide corroborative evidence rather than confirming the fall or health emergency.

These three metrics all contribute components into the final event score that once it exceeds a threshold and event is created and sent to the external software application. An event will contain essential information including but not limited to which microphone picked up the sound, timestamp of the event, the sound classification, and anomaly score/severity. This event will be sent over a wifi connection to the server backend application of the external software.

1.7.5 External Caregiver Application

The edge device will communicate with the external software, which is a standard web application following a three-tier client server architecture. It consists of a frontend UI to display the dashboard, backend to accept events and send email notification and a database to store events and account information.

1.7.5.1 Role base Access

To allow for multiple caregivers and for the care recipient to login they will each create their own accounts and link the respective device. This allows for multiple levels of access to a similar dashboard, avoids sharing credentials, and connects to the respective device. The minimum associated with a device is one caregiver and a care recipient but optional to add more caregivers access. The care recipient account is optional, they do not need an account but must consent to being monitored by the system during the setup process to add caregivers.

The notion of consent and passive monitoring is a complex area of legal and medical personal privacy. This capstone prototype will proceed under assumptions covered in section 1.12.2.

1.7.5.2 Frontend User Interface

The frontend user interface primarily displays a user friendly dashboard that will include device status, most recent event enlarged in main view with a summary, and event history list. The users will be able to manage their accounts and control the device from the application. Due to the role base access the dashboard from the care recipient view may redact information that may be sensitive. Sensitive information might include but is not limited to audio classification. It can be a vulnerable time when care systems are needed to continue living independently. They will also not be able to control the device i.e. mute or disable the microphones or turn off the device. From the caregiver's access and view and all controls are enabled and no information is redacted.

1.7.5.3 Backend and Notifications

The backend application will serve the frontend and accept events from the edge via REST APIs, will save incoming events to the database to display the history list of passed events. The business logic is very simple, no additional data manipulation is done, it only has to save new

events to the database, serve the front end and prepare new events for caregiver notification. This will reside on a cloud hosted platform.

Notification will be sent to all caregivers via the email addresses they used during sign up at account creation. This can be achieved at a small scale with Google Simple Mail Transfer Protocol server which is sufficient for the purpose of the system. Other means of notifying the caregivers of an event may be explored such as SMS. A challenge with SMS is implementation seems to cost monetarily even when considering workarounds to access the cell network directly.

1.7.5.4 Database

The database will essentially only store events and account information and configuration. There is not a lot of essential data that needs to be persisted. Additional data that could reside in the database is device status history, mute history, connection lost to edge device timestamps, basically system logs from the edge and web application. Some of the edge device logs may be available to the caregivers via the UI.

1.8 Comparison with Existing Solutions

The proposed system aims to address several limitations identified in the existing solutions reviewed in Section 1.2. Unlike Lifeline HomeSafe, the system does not require the resident to wear, charge, or manually activate a device. This reduces dependence on user compliance and allows monitoring to occur passively in the background. Lifeline provides reliable emergency communication, but its protection is reduced when the wearable is not being used [5].

Compared with camera-based systems such as the Nobi Smart Lamp, the proposed design does not use visual sensing. Instead, it will use microphones only, and all audio will be processed locally on the edge device. Only event data leaves the residence, which keeps personal information inside the home [6].

Radar-based solutions such as Vayyar Care and mmFall also provide passive, camera-free monitoring, but they rely on specialized radar hardware and complex signal-processing techniques. The proposed system will prioritize affordable, commercially available sensing and embedded-computing hardware. This will reduce cost and deployment complexity while still providing enough information for fall and anomaly detection [7], [9].

Compared with acoustic-based approaches such as Acoustic-FADE and Neo Care, the proposed system aims to combine fall detection with broader household behavioural anomaly detection using a learned acoustic baseline. While these existing systems demonstrate the potential of microphone-based monitoring, they also highlight challenges involving false alarms and reliable

detection in different environments. The proposed system will investigate whether combining acoustic event classification with contextual anomaly detection can help distinguish potentially dangerous situations from normal household activity. Local edge processing will also allow detection to continue without relying on continuous cloud connectivity [12], [13].

The proposed system also differs from solutions that focus exclusively on fall detection. Fall detection will remain a primary capability, while the learned acoustic baseline provides context around an event and supports detection of broader household anomalies. For example, a possible fall sound could be considered alongside the silence that follows it. This combination may help distinguish potentially dangerous situations from normal household events.

From a performance and reliability perspective, all detection will run locally on the edge device, which reduces latency and means the cloud only receives event data, never audio. The system is also intended to continue detecting events during a network outage, with alerts delivered once connectivity becomes available again.

The proposed system is not expected to outperform mature commercial products in every area. Its main advantage is the combination of passive operation, camera-free and wearable-free sensing, local edge processing, fall detection, and broader anomaly detection within a low-cost prototype. This approach aims to address privacy, usability, cost, and deployment limitations found across existing solutions.

1.9 Future Work / Project Phases

The project will be developed in stages, beginning with a focused prototype and leaving room for future expansion after the academic timeline. The short-term work will prioritize demonstrating the feasibility of the main system, while longer-term development could improve scalability, reliability, and practical deployment.

1.9.1 Short-Term Goals

Within the project timeline, the goal is to develop and validate a functional prototype of the proposed camera-free elder-care monitoring system. The project will progress through several phases:

- System design and microphone evaluation: Select suitable microphones and define the architecture.
- Prototype development: Integrate the microphones with a local edge-processing platform.
- Fall detection implementation: Develop and test fall-detection methods.
- Behavioral anomaly detection: Detect unusual activity patterns that may indicate risk.

- Alerting and monitoring: Notify caregivers when falls or high-risk anomalies are detected.
- System integration: Combine sensing, processing, detection, and notification components.
- Testing and evaluation**:** Measure detection accuracy, false alarms, missed events, and alert latency.

The primary objective within the project timeline is to demonstrate the feasibility of combining fall detection and behavioural anomaly detection in a passive, camera-free edge-AI system.

1.9.2 Long-Term Vision

Beyond the capstone project, the system could evolve into a more complete in-home monitoring platform for older adults living independently. Future versions could learn long-term behavioural patterns, adapt to changes in routine, and be evaluated in real residential environments with larger and more diverse datasets.

1.9.3 Scalability

The initial prototype will focus on one room monitored by a single microphone array. Future versions could scale to support:

- Multiple rooms and additional non-visual sensors.
- Multiple residents within the same household.
- Integration with existing smart-home infrastructure.
- Monitoring of multiple homes through a caregiver or care-provider platform.

A scalable architecture would allow new sensing and processing components to be added without major changes to the overall system.

1.9.4 Commercialization Potential

With further development and validation, the system could have potential as an assistive technology for older adults living independently. Possible users or customers could include families, home-care organizations, retirement communities, and healthcare providers.

A commercial version would require additional testing in areas such as reliability, safety, privacy, cybersecurity, regulatory compliance, and real-world performance.

1.9.5 Possible Extensions and Improvements

Potential future improvements include:

- More advanced and personalized fall/anomaly detection models.
- Combining multiple sensor types to improve detection reliability.
- Finer-grained severity levels learned from longer-term household data.
- A caregiver-facing mobile application.
- Integration with healthcare or emergency-response systems.
- Reduced hardware cost and installation complexity.

1.10 Risk Analysis & Mitigation

Several technical, resource, and schedule risks may affect the successful development of the prototype. Table 1.3 identifies the major project risks, evaluates their likelihood and impact, and outlines mitigation strategies intended to reduce their effect on the project.

Risk	Likelihood	Impact	Mitigation
False Negative Detection	**Medium**	**High**	- Test across different fall sounds. - Test across different distances. - Test amongst different background noises. - Test among different detection thresholds.
False Positive Detection	**High**	**Medium**	- Include non fall sounds during testing.
Model performs well on SAFE, but poorly in real situation	**Medium**	**High**	- Add controlled recordings in different environments and note that results show prototype performance,

			not real-world accuracy.
Cannot run model with acceptable latency	**Medium**	**High**	- Use lightweight models suitable for embedded hardware. - Benchmark inference latency on the target device and simplify the model or processing pipeline if necessary..
Network outage prevents the caregivers notification	**Medium**	**High**	- Keep detection running offline and send alerts once connection returns.
Component unavailable or over budget	**Medium**	**Medium**	- Use affordable and widely available parts. - Have backup options. - Use versatile components.
Development is behind schedule	**Medium**	**High**	- Prioritize fall detection first.
Insufficient or unrepresentative	**Medium**	**High**	- Learn the baseline from the week of

data for learning normal household behaviour			continuous home audio in the SINS database. - Test anomaly detection across different days and rooms in the dataset. - Clearly report limitations of the learned baseline.

Table 1.3: *Project Risks and Mitigation Strategies*

1.11 Health & Safety Considerations

1.11.1 Electrical safety

The prototype design will utilize low-voltage embedded computing and sensing electronics. Components will need to be operated within their manufacturer specifications of voltage and currents and powered by an appropriate power supply. The device needs to be housed, where feasible, in order to avoid any unintentional contact with electrical conductors. Damaged cables and components should not be utilized.

1.11.2 Thermal risks

Extended edge-AI processing can result in heat generation on the embedded processor and its associated electronics. As such, thermal behavior needs to be monitored. Appropriate ventilation and manufacturer recommended cooling must be provided, and device installation should avoid flammable material or locations where ventilation vents can become blocked.

1.11.3 Mechanical hazards

Microphone placement, processing electronics, power cables, and other component placement should occur in such a way that there will be no falling or tripping hazards created. The prototype design should not interfere with pathways in a way which would be hazardous, especially since the target use case includes older adults who are potentially at risk of falls.

1.11.4 User safety

The prototype is designed to alert users to safety issues and should not replace the need for emergency services and professional monitoring. Failure to detect events can lead to delays in getting help, whereas false alarms can lead to unwarranted anxiety of care-providers. The prototype will not automatically contact 911 but will escalate an event to the user or another human contact for verification.

1.12 Ethical Considerations

1.12.1 Privacy and data security

The presence of continuous microphones causes issues regarding the privacy of the resident because household audio can contain private conversations and sensitive information. The proposed design is based on the principle of data minimization: raw audio is processed and stored only on the edge device for a limited period of time defined in FR2 and is never sent anywhere else. Only event information including the event type, room, timestamp, and time elapsed since the start of the event is sent. A physical mute button is also implemented which makes it possible for the resident to turn off audio recording. That approach is well supported by the Canadian research-ethics guidance. According to TCPS 2, privacy and confidentiality should be taken into account throughout the entire lifecycle of the data and the recommended safeguards should depend on the sensitivity of the information. [23]

1.12.2 Consent and Autonomy

Monitoring should take place only after obtaining informed consent of people being monitored. Residents should know what data is collected, how it is processed, what data is sent from the device, and who gets alerted. There is also a problem with visitors and other household members because their sounds can be recorded even though they are not the target user.

The notion of consent and passive monitoring introduces complex legal, ethical, and medical privacy considerations. For the purposes of this capstone prototype, the following assumptions will be made:

- **Capacity to consent:** The care recipient has the mental capacity and physical ability to provide informed and voluntary consent to the monitoring system.
- **Initial consent:** During the initial setup process, the care recipient provides informed consent to acoustic monitoring and authorizes the initial caregiver to access the system's monitoring data.

- **Trusted caregiver:** Following initial consent, the initial caregiver is designated as a trusted user with administrative privileges. This user may authorize additional caregiver accounts with read-only access to monitoring data without requiring repeated consent from the care recipient.
- **Scope of authorization:** Additional caregivers cannot modify monitoring configurations, manage other accounts, or grant access to additional users. These privileges remain restricted to the trusted caregiver.
- **Revocation of consent:** The care recipient retains the ability to withdraw consent to monitoring, at which point monitoring and subsequent data collection must cease.

1.12.3 Bias in algorithms (if AI/ML involved)

The performance of AI may be affected by differences in flooring, acoustics, noise, mobility patterns, features of falls, household activities of people who have not been included into the training data set. For example, testing on SAFE or controlled experiments will not provide information about the system's performance on all environments and people. That is why testing should cover different acoustic conditions and the limitations of the dataset should be mentioned explicitly.

1.12.4 Environmental impact

Environmental impact should be reduced through low-power edge computing, avoiding unnecessary sensors, reusing the development hardware when possible and proper disposal of the electronic components at the end of the project. Edge computing also excludes transmitting raw audio to cloud infrastructure, although the prototype itself will continuously consume some electricity.

1.12.5 Responsible engineering practices

The prototype should not be presented as a clinically validated emergency-response system or a medical diagnostic tool. Accuracy, false-positive rate, false-negative rate, and latency should be provided clearly. The system should assist caregivers rather than give them a feeling of security.

1.13 Relevance to Degree

The project team consists of five students enrolled in the Software Engineering program within the Department of Systems and Computer Engineering at Carleton University. The proposed Edge-AI Home Anomaly Detection system incorporates several key areas of software engineering, including embedded systems, software architecture, real-time processing, software

validation, and project management. Knowledge and skills developed throughout the program will be applied to the design, implementation, and evaluation of the prototype.

The embedded hardware and sensing components of the project relate closely to SYSC 3310, “Introduction to Real-Time Systems”, which covers embedded programming, hardware interfacing, input/output methods, interrupts, and real-time constraints. These concepts will be relevant when integrating sensors with the edge-processing device and ensuring timely data acquisition. Additionally, SYSC 3303, “Real-Time Concurrent Systems”, provides knowledge of concurrent processing and synchronization, which will be applied to the management of continuous sensor input, anomaly detection, and alert generation.

The software architecture and integration aspects of the project build on concepts from SYSC 4120, “Software Architecture and Design”, particularly the organization of software into modular components and the design of interactions between them. SYSC 4806, “Software Engineering Lab”, further supports the development of the monitoring and alerting components through its emphasis on full-stack development, collaborative programming, code reviews, and system integration.

Testing and evaluating the system will draw on SYSC 4101, “Software Validation”, and ECOR 2050, “Design and Analysis of Engineering Experiments”. Software validation techniques will help establish systematic testing procedures, while statistical and experimental methods will support the evaluation of detection accuracy, false-positive and false-negative rates, and system performance under different environmental conditions.

Finally, SYSC 4106, “The Software Economy and Project Management”, provides a foundation for managing project resources, schedules, risks, and development priorities. These concepts are particularly important given the limited budget, academic timeline, and technical challenges associated with developing a functional prototype.

The project provides an opportunity to integrate theoretical knowledge and practical engineering skills acquired throughout the Software Engineering program into the development of a complete, edge-AI monitoring system.

1.14 Task Distribution

The project will be developed collaboratively by five team members, with work divided across the major components of the proposed system. Table 1.4 outlines the preliminary distribution of responsibilities, including the main tasks and expected deliverables for each area. Specific assignments may be revised as the architecture and technical requirements become clearer.

Work Area	Responsibilities	Expected Deliverables	Primary Member	Secondary Member
Hardware and Edge Integration	Evaluate and integrate suitable microphones, configure the edge-processing device, implement audio data acquisition, and establish communication between hardware and software components.	Functional microphone-based sensing setup and edge-processing platform capable of capturing and processing audio data locally.	Matthew Ross	Ihsan Aydin
Data Collection and Preprocessing	Identify suitable audio datasets, collect representative normal household and simulated fall sounds, implement audio preprocessing and feature extraction, and prepare data for model training and evaluation.	Prepare the SAFE and SINS datasets, record staged fall sounds, implement audio preprocessing and leverage pretrained YAMNet for feature extraction. Prepare labelled embeddings for classifier training.	Fareed El Bokl	Yannis Barasa
Specialized Event Classification	Train a purpose specific audio classifier following the Tensor flow YAMNet Transfer learning tutorial with the SAFE data set[20]. Evaluate detection accuracy under different conditions, and prepare for edge deployment.	Specialized audio classification with documented performance results, integrated with the edge-processing system.	Yannis Barasa	Fareed El Bokl
Acoustic and Behavioural Anomaly Detection	Develop a statistical household acoustic baseline using YAMNet embeddings. Investigate Mahalanobis-distance anomaly scoring using mean embeddings and covariance statistics. Evaluate alternative methods and establish anomaly thresholds. Investigate baseline calibration, adaptation,	Baseline management and acoustic anomaly detection components producing anomaly scores. Evaluation of baseline representation, anomaly detection performance, false alerts, and potential adaptation strategies.	Ihsan Aydin	Matthew Ross

Work Area	Responsibilities	Expected Deliverables	Primary Member	Secondary Member
	and temporal household acoustic activity patterns.			
Alerting and Monitoring Interface	Develop the event-handling and notification mechanisms, implement the caregiver web dashboard, and establish communication between the edge device and monitoring interface.	Functional alerting mechanism and web interface displaying detected events and system status.	Samy Kaddour	Ihsan Aydin

Table 1.4: *Preliminary Task Distribution*

Although each work area will have a primary and secondary contributor, system integration, testing, documentation, and project planning will be shared responsibilities. Progress will be reviewed through weekly meetings and regular communication, and assignments may be adjusted based on workload, technical needs, and individual strengths.

1.15 Schedule & Milestones

The overall project has been divided into four main phases that follow the established timeline imposed by the department. The table 1.5 and gantt chart figure 1.3 below outline the start and end date of each phase. The end date of a phase is often a predetermined date that falls on a submission of a deliverable and the start date is the following day.

Phase	Start Date	End Date
Phase 1: Proposal	September 9th, 2026	October 23rd, 2026
Phase 2: Progress Report + Development	October 24th, 2026	December 11th, 2026
Phase 3: System Testing + Presentation	December 12th, 2026	January 31st, 2027
Phase 4: Final Report + Poster Fair	February 1st, 2027	April 9th, 2027

Table 1.5: *Project Phase Start and End date outline*

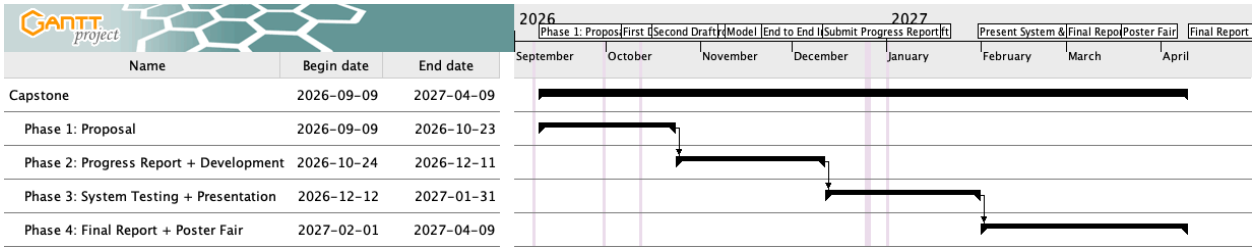


Figure 1.3: *Capstone Project Gantt Chart*

1.15.1 Phase 1: Proposal

Phase 1 runs from September 9th, 2026 to October 23rd, 2026 and covers the project Proposal period. The tasks are derived from what needs to be included in the Project Proposal. Milestones are depicted on the Gantt Chart by black diamonds. The milestones in Phase 1 are the first draft of the Proposal, second draft, and the Proposal submission as they are the major indicator of progress through this phase.

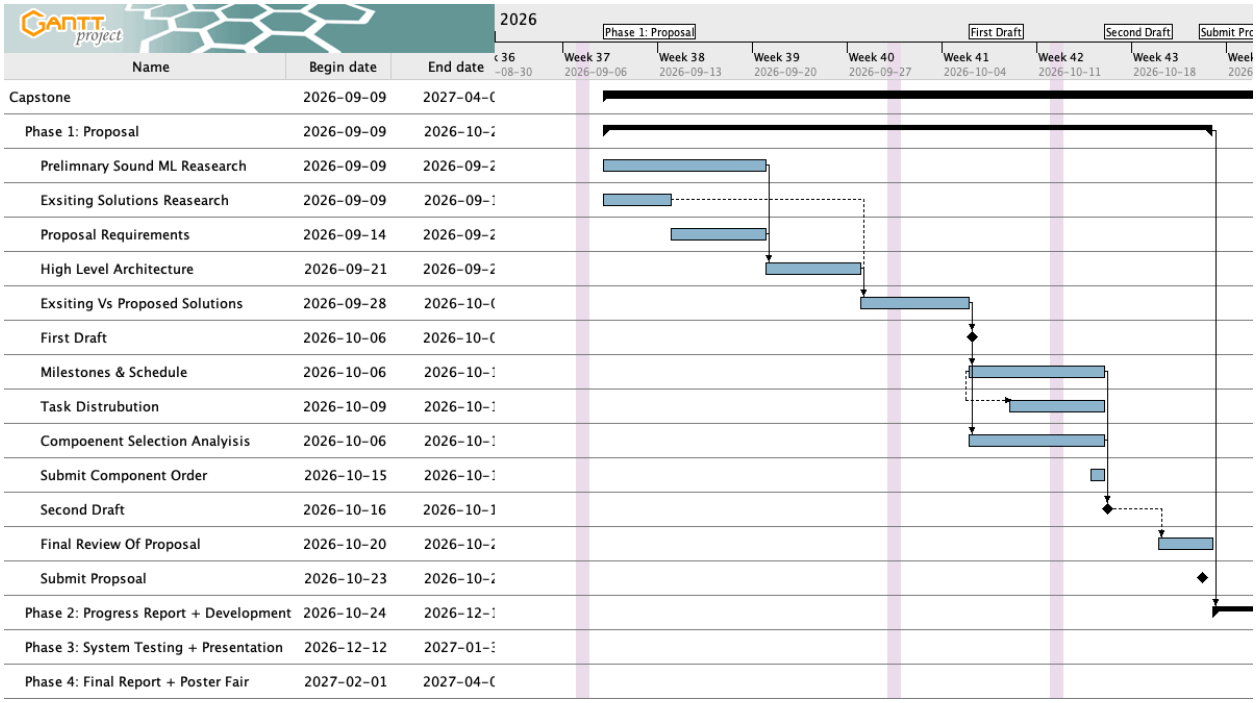
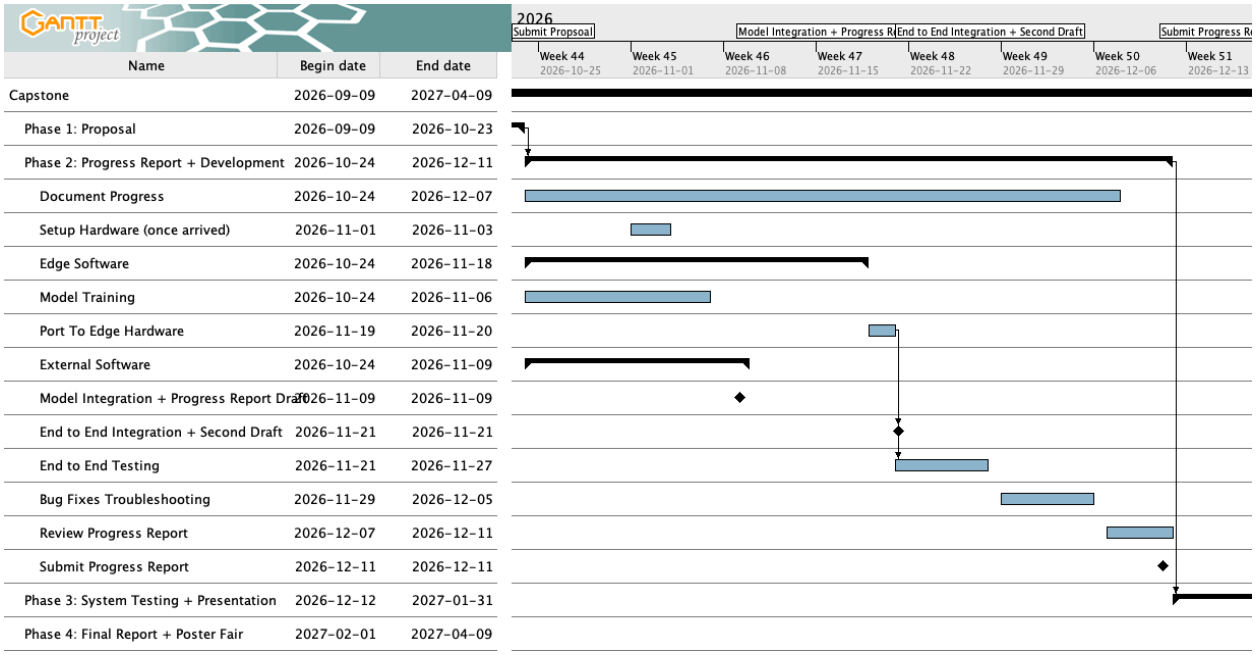


Figure 1.4: *Capstone Phase 1: Proposal Gantt Chart*

1.15.2 Phase 2: Progress Report + Development

Phase 2 is the bulk of the work, this phase encapsulates building the system and documenting the progress in the progress report. This phase runs from October 24th, 2026 to December 11th 2026 from reading week till the last day of classes in the fall term. The milestones in Phase 2 are: Model integration and first draft, porting to edge hardware and second draft, and submission of the Progress Report.



Name	Begin date	End date
Capstone	2026-09-09	2027-04-09
Phase 1: Proposal	2026-09-09	2026-10-23
Phase 2: Progress Report + Development	2026-10-24	2026-12-11
Document Progress	2026-10-24	2026-12-07
Setup Hardware (once arrived)	2026-11-01	2026-11-03
Edge Software	2026-10-24	2026-11-18
Model Training	2026-10-24	2026-11-06
Port To Edge Hardware	2026-11-19	2026-11-20
External Software	2026-10-24	2026-11-09
Model Integration + Progress Report Draft	2026-11-09	2026-11-09
End to End Integration + Second Draft	2026-11-21	2026-11-21
End to End Testing	2026-11-21	2026-11-27
Bug Fixes Troubleshooting	2026-11-29	2026-12-05
Review Progress Report	2026-12-07	2026-12-11
Submit Progress Report	2026-12-11	2026-12-11
Phase 3: System Testing + Presentation	2026-12-12	2027-01-31
Phase 4: Final Report + Poster Fair	2027-02-01	2027-04-09

Figure 1.5: *Collapsed View of Phase 2 Gantt Chart*

Phase 2 is organized by the major software development of the system and documenting these steps. There are two major software development aspects: the edge software including the model and the external software web app. Above figure 1.5 is a collapsed view of Phase 2 it depicts the entirety of the phase while hiding development tasks collapsed under the Edge software and External software depicted by the black period bars.

Figure 1.6 below provides an expanded view of the development tasks required for the software running on the edge device. This includes implementing the audio pipeline, which involves capturing audio streams from the microphones, preprocessing the audio, and extracting acoustic features for input into the model. Once the audio pipeline is established, model integration and model output handling can be developed in parallel. After the model has been integrated and its

outputs are handled correctly, the edge software can be connected to the external application through the communication interface.

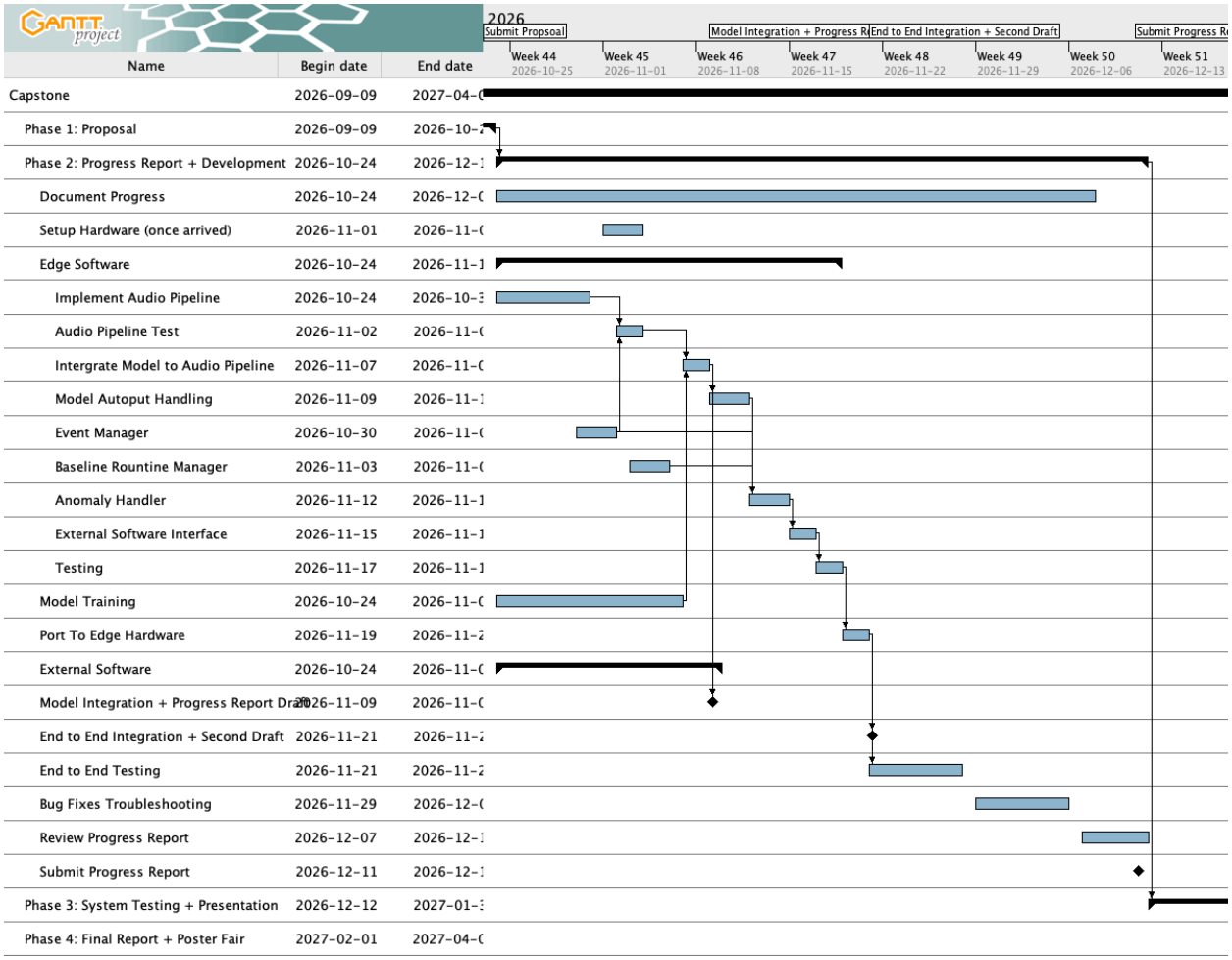
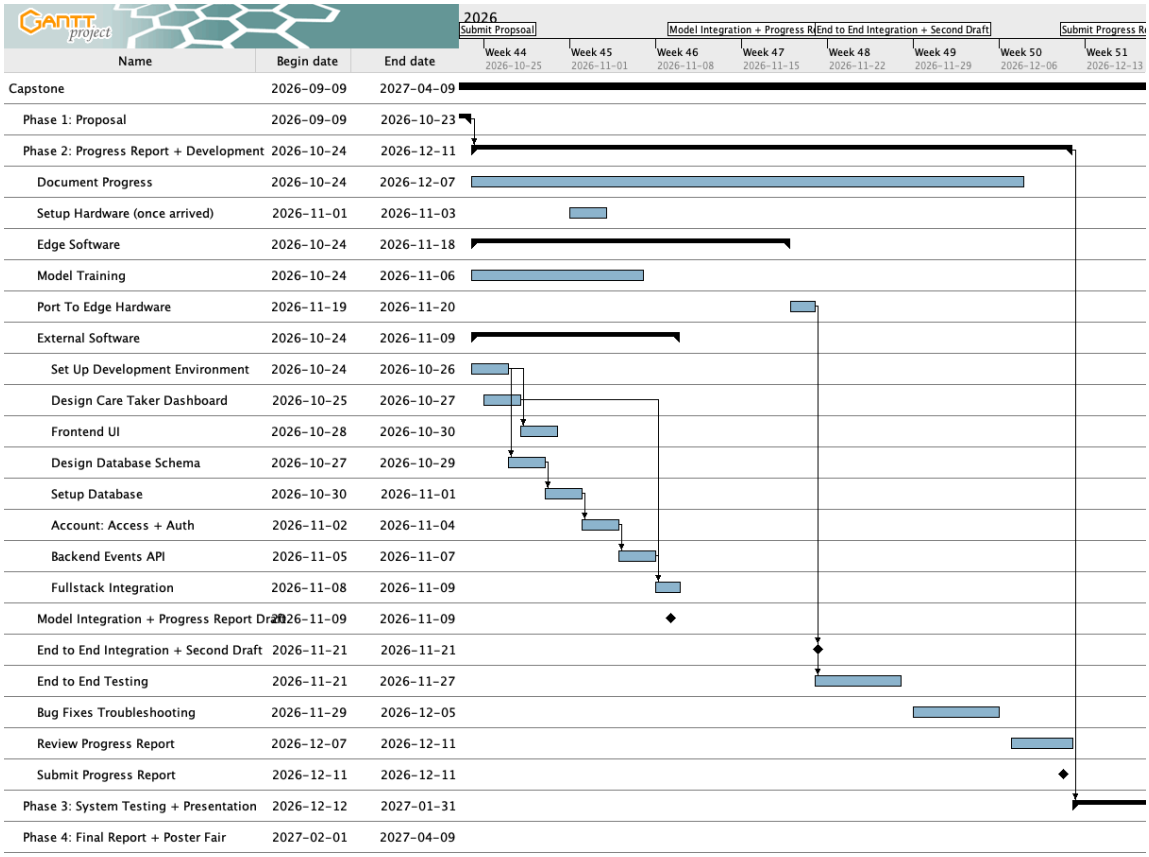


Figure 1.6: *Edge Software Expanded View of Phase 2 Gantt Chart*

Lastly, figure 1.7 is the expanded view of the External Software portion of the system. This is a standard web application that will have a backend and frontend and database. Logically, the development environment is set up and the UI design is done early to establish the best user experience for the caretaker. The database schema is also established early to identify what kind of data is stored, handled and displayed to the user. Once the database is set up, access and authentication can be implemented to access your account to view the dashboard. The Backend API is essential and is the component that will allow for communication between the Edge and External to notify care takers.

At this moment once the edge is functional and the web app is deployed end to end testing can begin. End to end integration can be done and testing can commence. During this period labeled audio will be used at first to ensure the system is working as intended and gradually unlabel

audio will be tested. The system will be tested in scenarios approximating its intended use by us members. There is an intentional long period for bug fixes and troubleshooting in case the issues are uncovered or any of the previous tasks shift to the right. Lastly the Progress report is reviewed and submitted ending the second phase.



Name	Begin date	End date
Capstone	2026-09-09	2027-04-09
Phase 1: Proposal	2026-09-09	2026-10-23
Phase 2: Progress Report + Development	2026-10-24	2026-12-11
Document Progress	2026-10-24	2026-12-07
Setup Hardware (once arrived)	2026-11-01	2026-11-03
Edge Software	2026-10-24	2026-11-18
Model Training	2026-10-24	2026-11-06
Port To Edge Hardware	2026-11-19	2026-11-20
External Software	2026-10-24	2026-11-09
Set Up Development Environment	2026-10-24	2026-10-26
Design Care Taker Dashboard	2026-10-25	2026-10-27
Frontend UI	2026-10-28	2026-10-30
Design Database Schema	2026-10-27	2026-10-29
Setup Database	2026-10-30	2026-11-01
Account: Access + Auth	2026-11-02	2026-11-04
Backend Events API	2026-11-05	2026-11-07
Fullstack Integration	2026-11-08	2026-11-09
Model Integration + Progress Report Dra	2026-11-09	2026-11-09
End to End Integration + Second Draft	2026-11-21	2026-11-21
End to End Testing	2026-11-21	2026-11-27
Bug Fixes Troubleshooting	2026-11-29	2026-12-05
Review Progress Report	2026-12-07	2026-12-11
Submit Progress Report	2026-12-11	2026-12-11
Phase 3: System Testing + Presentation	2026-12-12	2027-01-31
Phase 4: Final Report + Poster Fair	2027-02-01	2027-04-09

***Figure 1.7**: External Software Expanded View of Phase 2 Gantt Chart*

1.15.3 Phase 3: System Testing + Presentation

Phase 3 at this point in the Capstone project is envisioned to have a functional system that can begin collecting data for the Final Report and presentation. This phase runs from December 12th, 2026 to the end of the first month of the new year January 31st, 2027. As this phase is over the exam and holiday period the tasks stretch to accommodate for this time of year this can be seen in figure 1.8 below. The only milestone is the Oral presentation some time in the last week. The majority of the phase is collecting data, analyzing the collected data and preparing for the presentation at a date to be decided by the department.

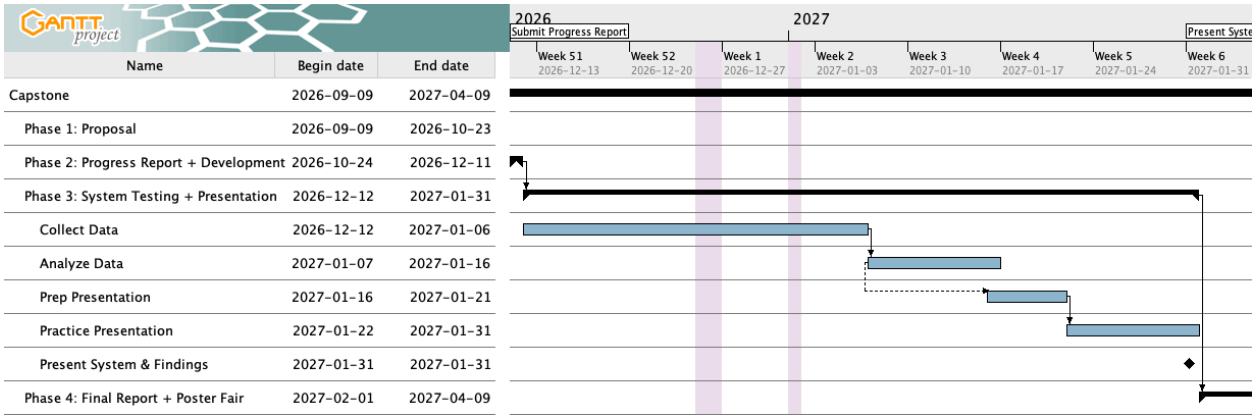


Figure 1.8: *Phase 3 System Testing + Presentation Gantt Chart*

1.15.4 Phase 4 Final Report + Poster Fair

Phase 4 is the last phase of the Capstone project that will wrap up the last six months of work with the final report and the poster fair. This phase period runs from after the oral presentation in Phase 3 plan to begin February 1st, 2027 and run till the last day of classes in the second semester April 9th. The majority of the phase is writing the final report and preparing for the poster fair. The milestones in this phase are the final report draft, the poster fair, and the final report submission as seen below in figure 1.9.

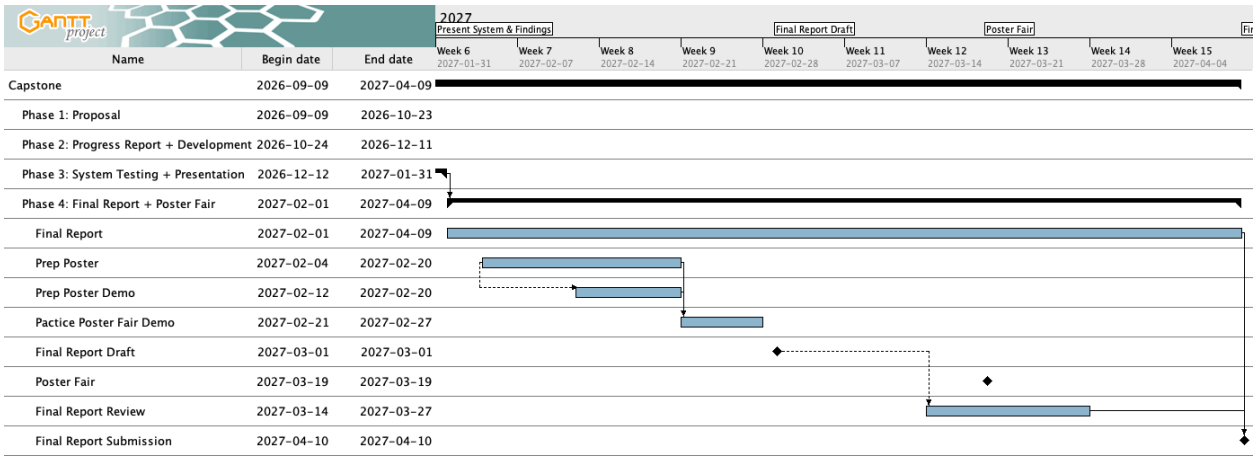


Figure 1.9: *Phase 4 Final Report + Poster Fair Gantt Chart*

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